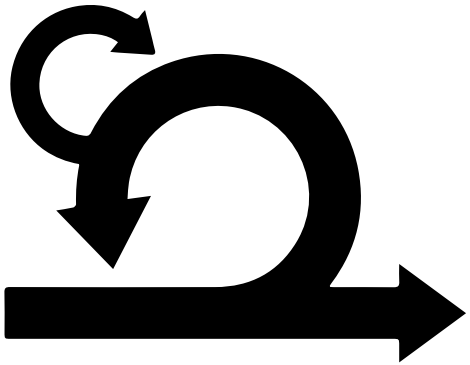
System Design

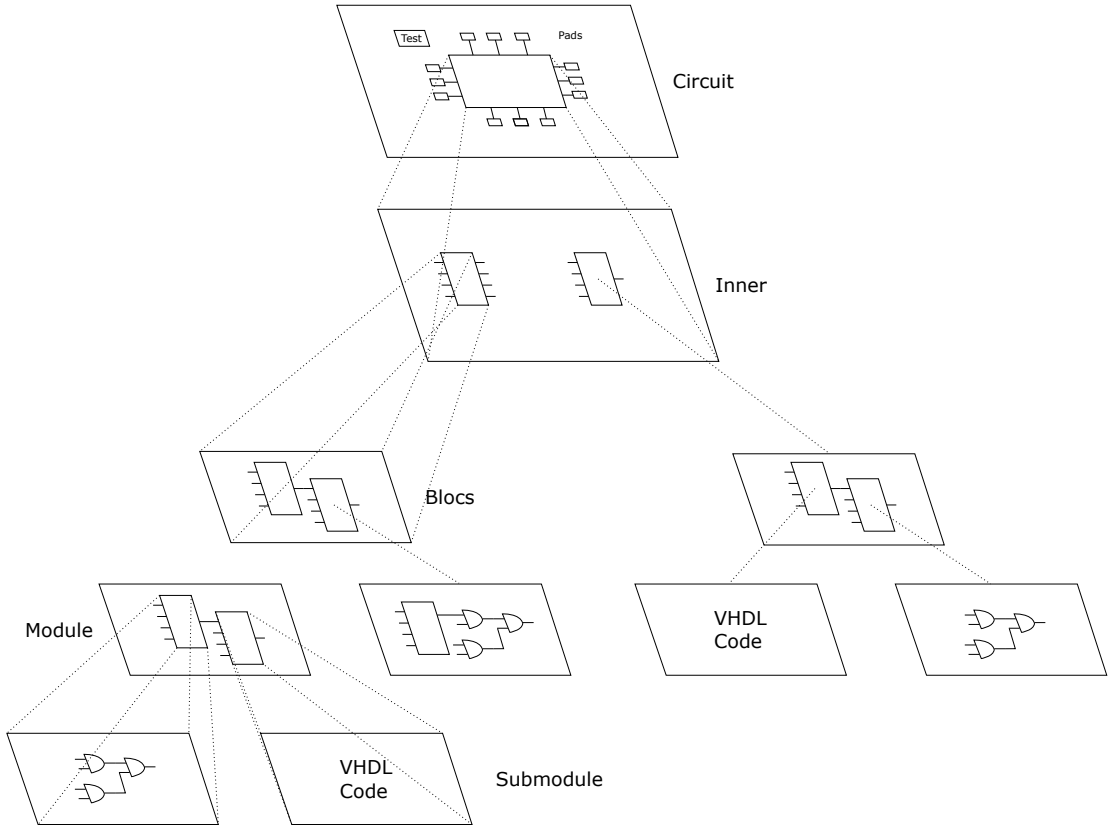
Technical Management



Systems Engineering program

Silvan Zahno silvan.zahno@hevs.ch

What for and why?



Technical Management Methods

PRINCE2

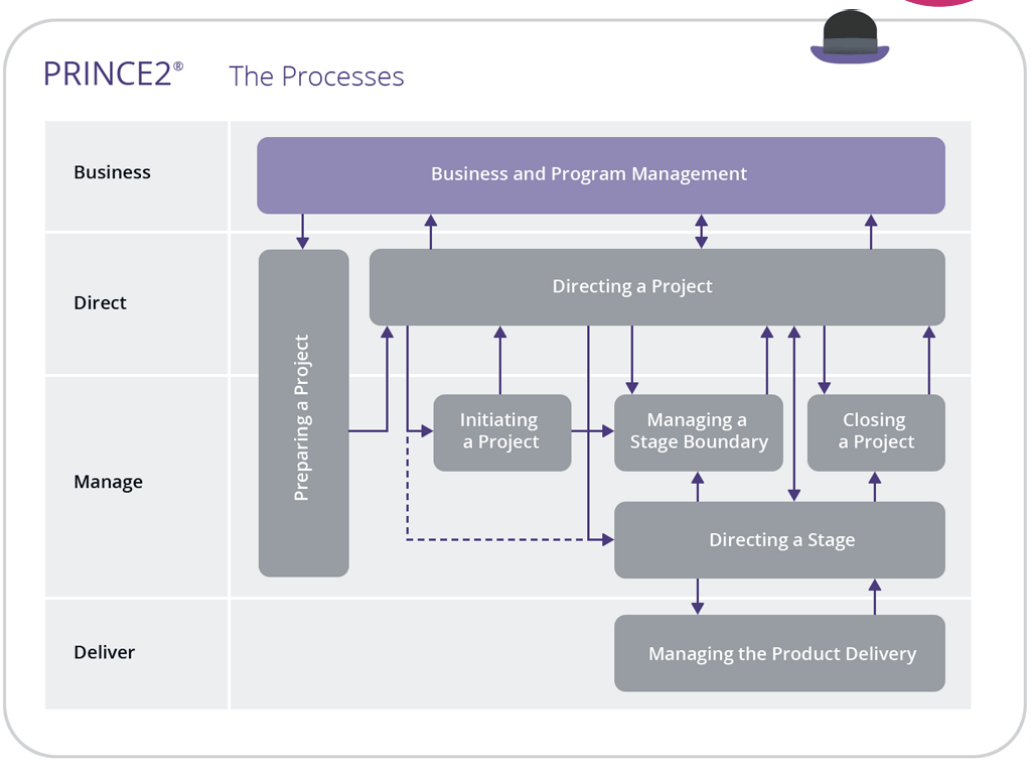
Organized approach to achieving project goals within constraints.

What it is:

Used for quality management to eliminate project defects.

Who should use it:

Large teams that tackle complex projects



https://www.microtool.de/en/knowledge-base/how-does-prince2-work/

Technical Management Methods

Six Sigma

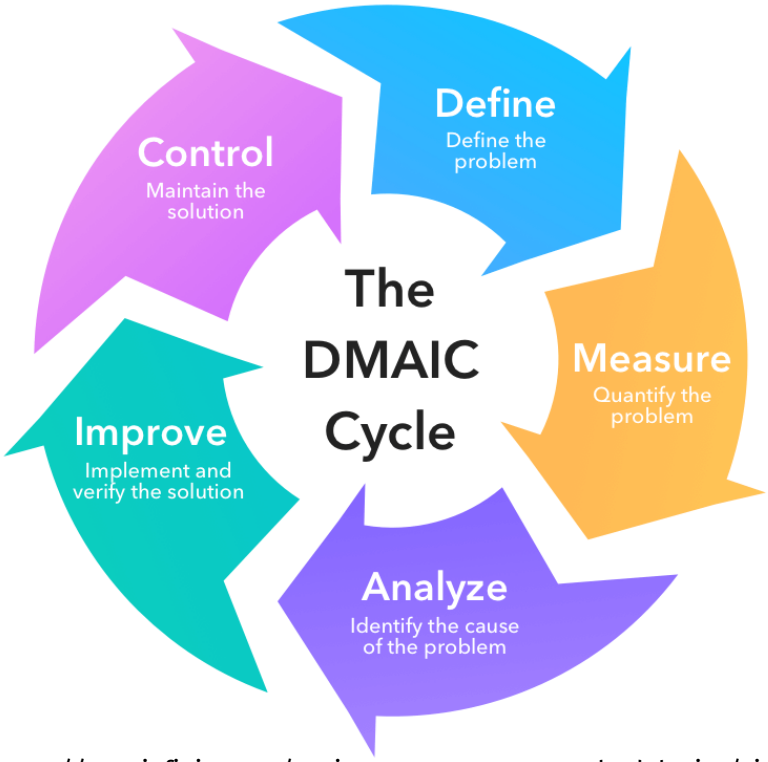
Six Sigma is a method for improving process quality by reducing variability and eliminating defects.

What it is:

Used for quality management to eliminate project defects.

Who should use it:

Large organizations looking to reduce errors



https://startinfinity.com/project-management-methodologies/six-sigma

Technical Management Methods

Lean (Production)

Lean is a methodology for optimizing process efficiency and reducing waste by focusing on adding value from the customer's perspective and continuously improving the process flow.

What it is:

Aims to cut waste and create a simple framework for project needs

Who should use it:

Teams struggeling with efficiency issues

https://www.sketchbubble.com/en/presentation-lean-project-management.html

Technical Management Methods

Extreme Programming (XP)

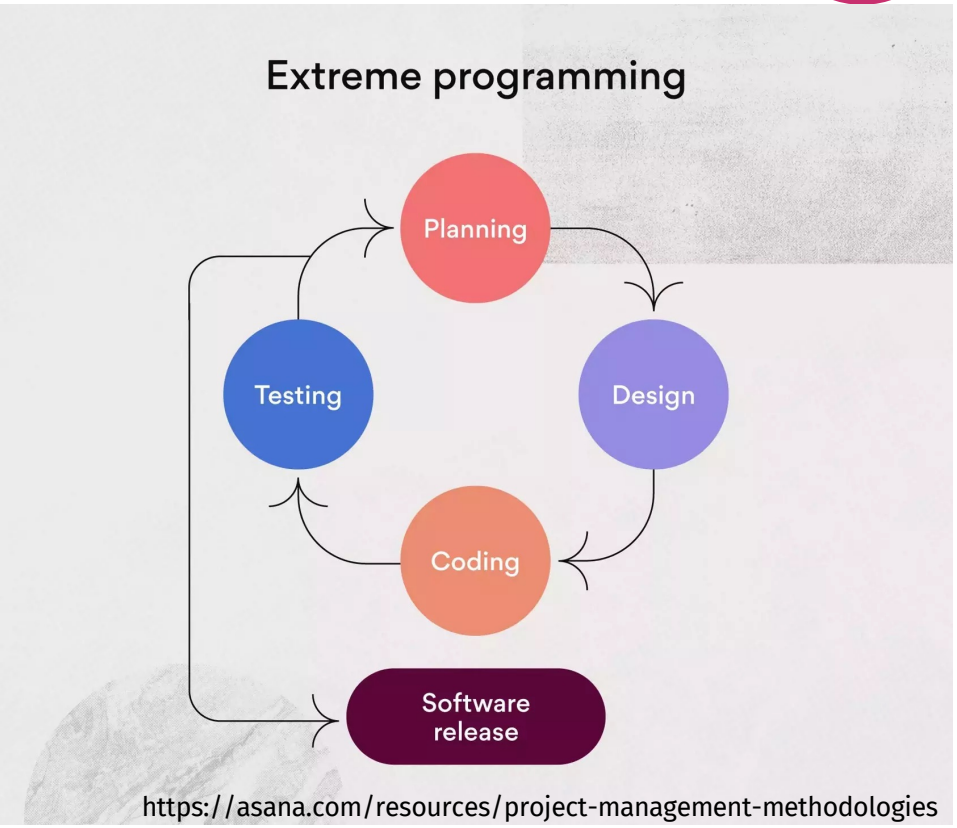
Extreme Programming (XP) is an agile software development approach that focuses on frequent releases, continuous testing, and customer feedback to deliver high-quality software products.

What it is:

Works by creating short development cycles with many releases

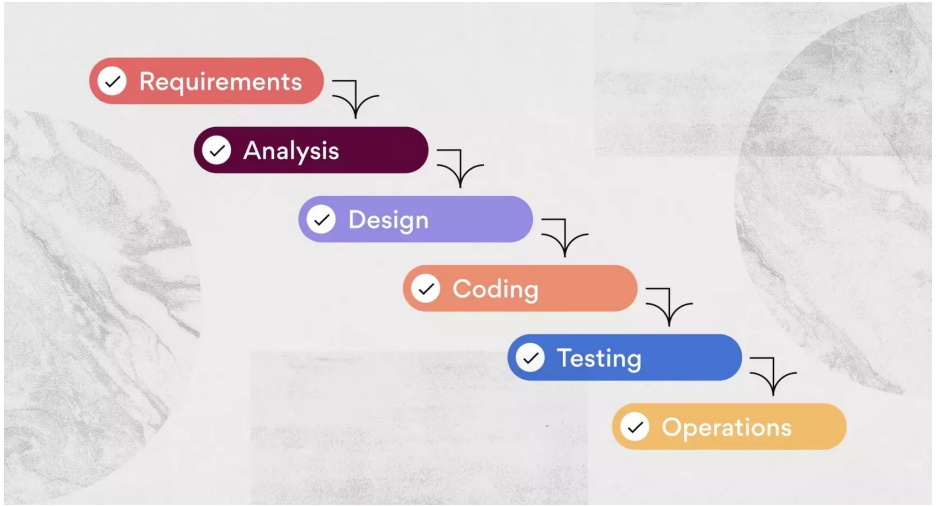
Who should use it:

Small top mid-size teams that have tight turnaround times



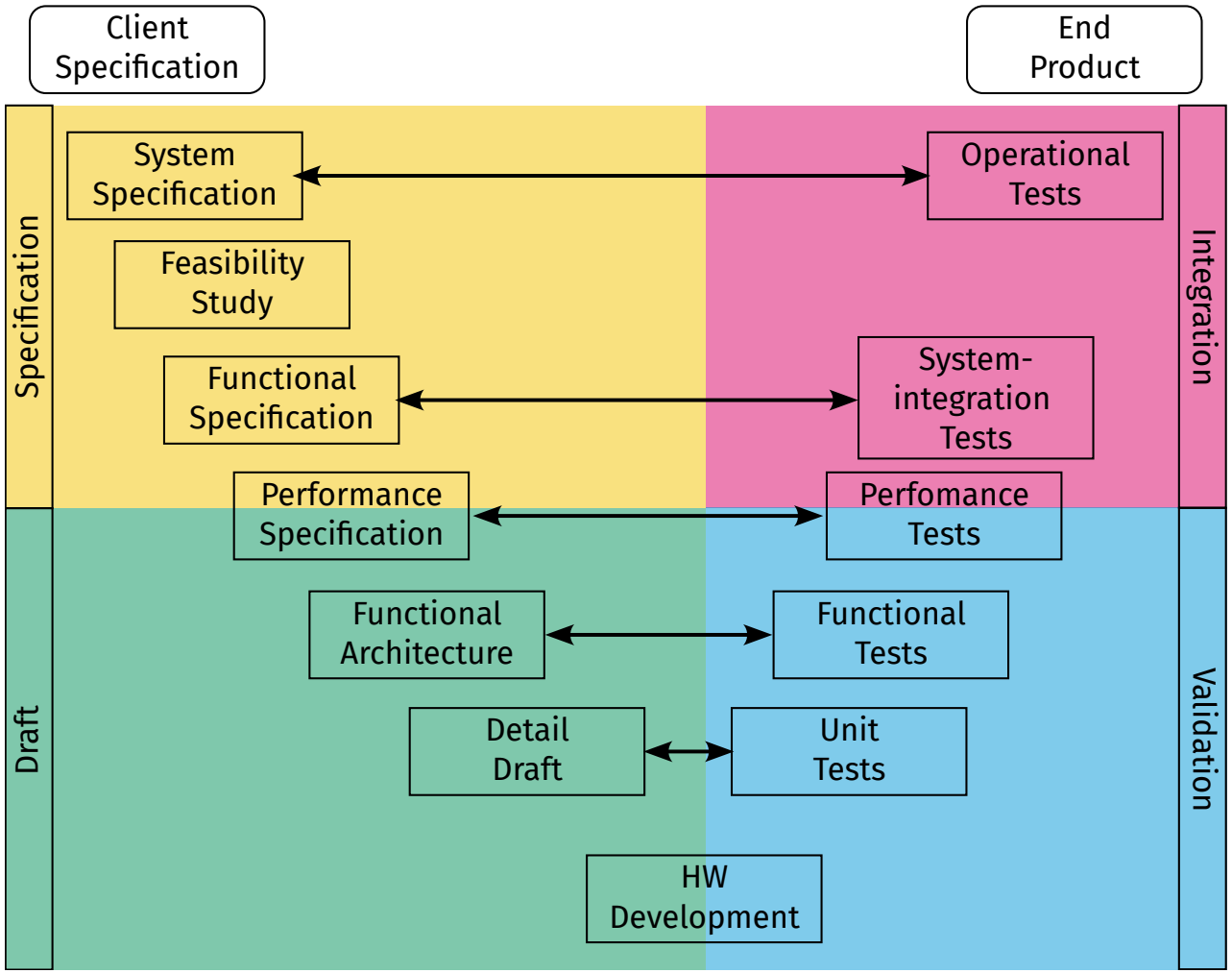
https://asana.com/resources/project-management-methodologies

Waterfall / Diagram-V



https://asana.com/resources/project-management-methodologies

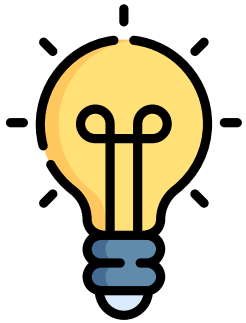
Diagram-V


Client Specification
End Product
Specification
Draft
Integration
Validation
System Specification
Operational Tests
Feasibility Study
Functional Specification
System-integration Tests
Performance Specification
Perfomance Tests
Functional Architecture
Functional Tests
Detail Draft
Unit Tests
HW Development

Agile

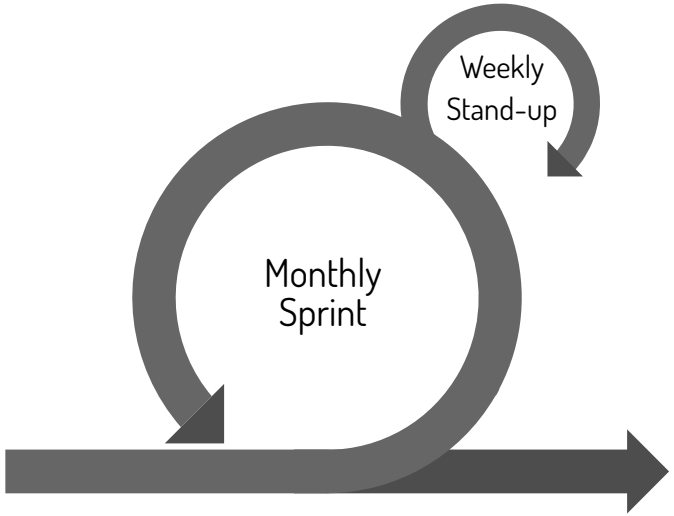
Scrum / Kanban

Scrum

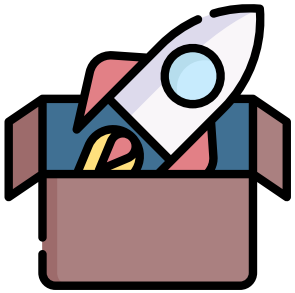


IDEA / VISION
Business Model
Milestones
Releases

PRODUCT BACKLOG
Project specific
prioritized workpackages



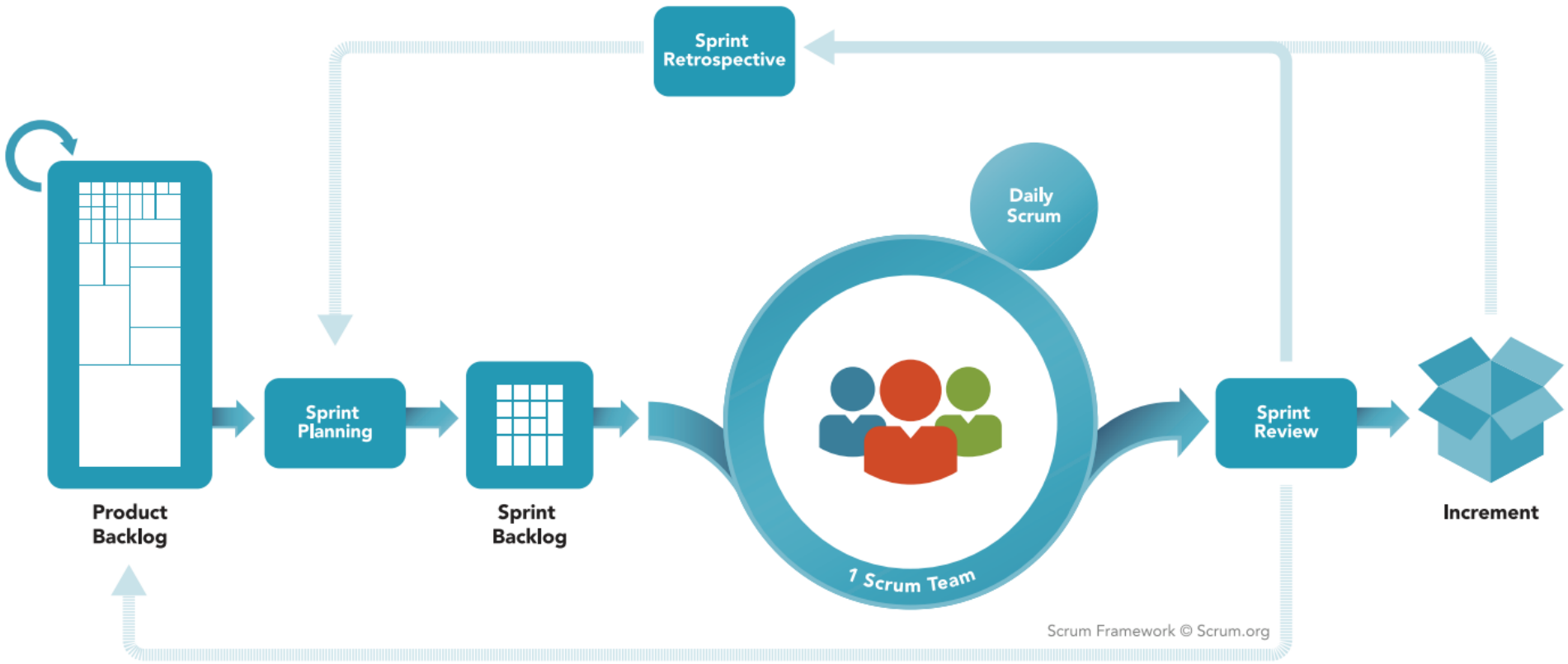
SPRINT PLANNING
Sprint Retrospective
Next Sprint Planning



FINISHED PRODUCT

Icons made by Freepik from Flaticon

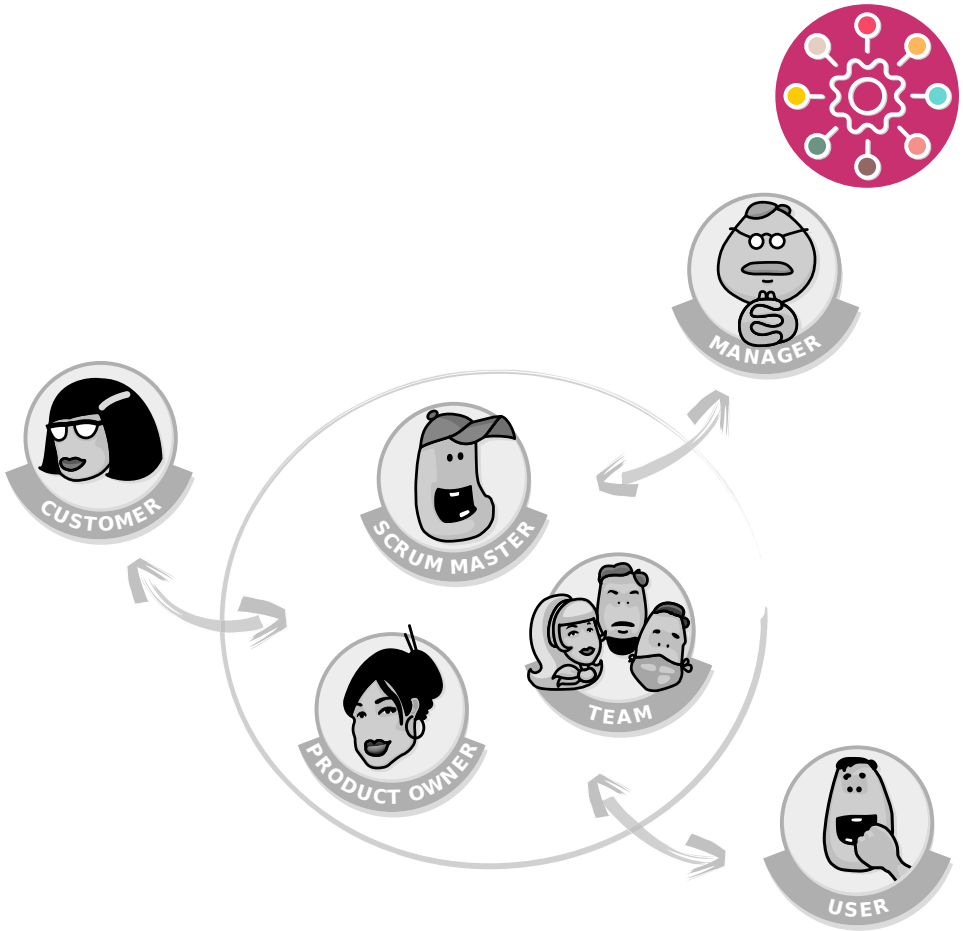
Scrum


Sprint Retrospective
Daily Scrum
Product Backlog
Sprint Planning
Sprint Backlog
1 Scrum Team
Sprint Review
Increment
Scrum Framework © Scrum.org

Practical Scrum

- The Roles in the Scrum Team
  - Product Owner
  - Development Team
  - Scrum Master

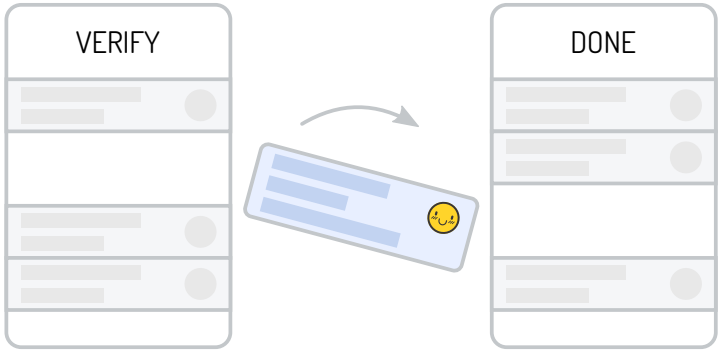
- The Scrum artefacts
  - User Story
  - Product Backlog
  - Sprint Backlog
  - Increment
  - Definition of Done, Definition of Ready



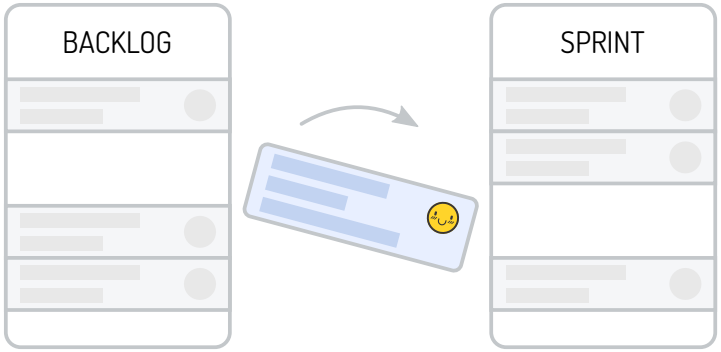
B. Gloger and K. Schwaber, *Scrum: Produkte zuverlässig und schnell entwickeln*, 4., Überarb. Aufl. München: Hanser, 2013.

Scrum Events

Sprint Retrospective



Sprint Planning



User Story

Explicit Title

- e.g. for a Taxi ride app: "Search for an available Taxi"

Narrative phrase: "As [persona], I want [feature], so that [function"]

- e.g. "As a customer, I want to see all nearby taxis, so that I can select one"

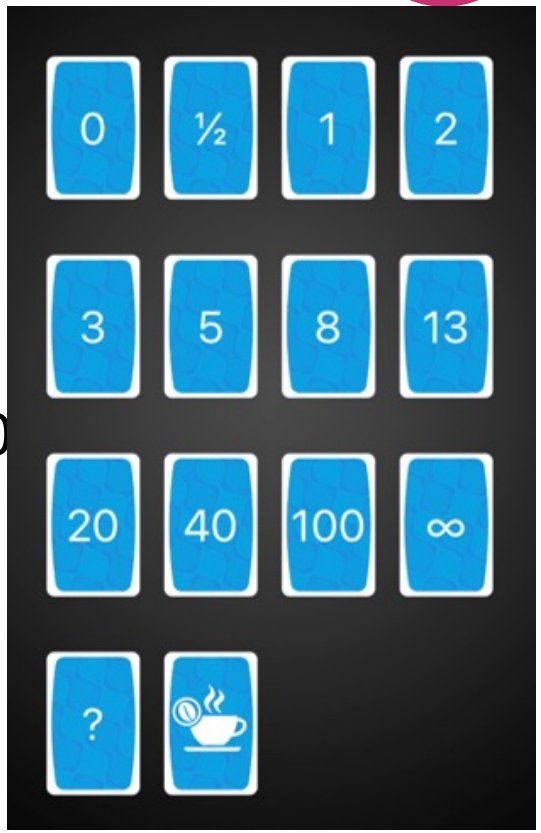
Acceptance Criteria

- e.g. "Only display available taxis"
- e.g. "Display them on a list from the closest to the furthest away"
- e.g. "Show a picture of the car"

Task Estimation

Scrum Time App

- Within the team 1 points needs to be defined and refined.
- Sequences
  - Planning Poker: 0, ½, 1, 2, 3, 5, 8, 13, 20, ..., 100
  - Fibonacci: 0, 1, 2, 3, 5, 8, 13, 21, 144
  - Natural: 1, 2, 3, 4, 5, 6, ..., 16
  - T-Shirt: XS, S, M, L, XL

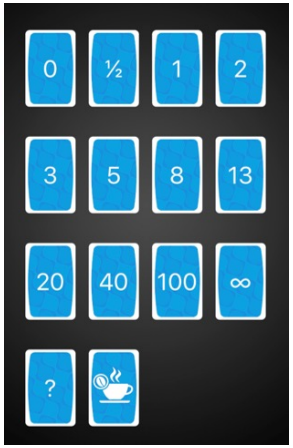


Task Estimation

Example 1

Search a Lego in a Pile and stack is corresponds to:

1/2 Point

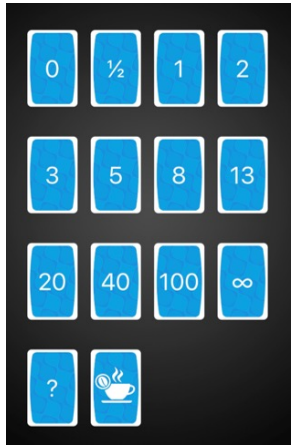


Task Estimation

Example 2

Create a small Lego robot

5 Points

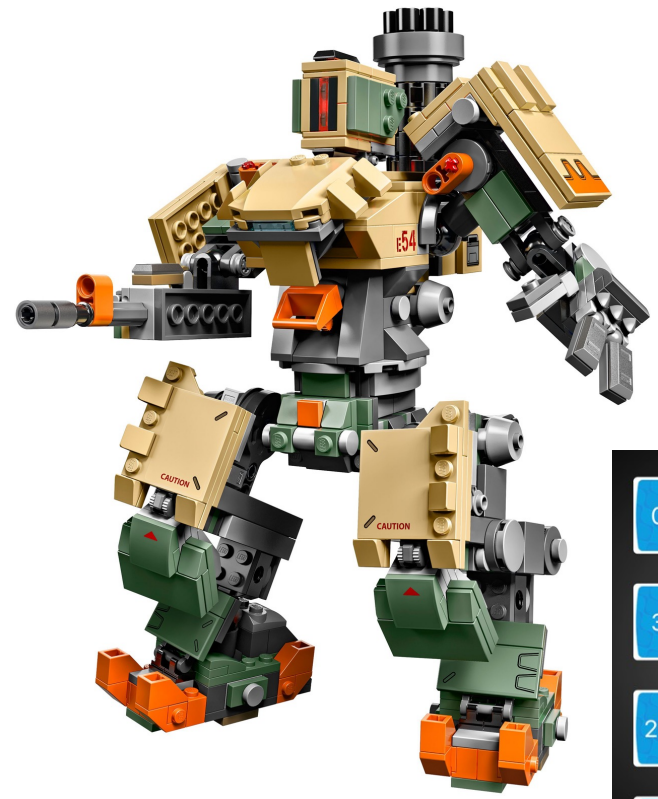


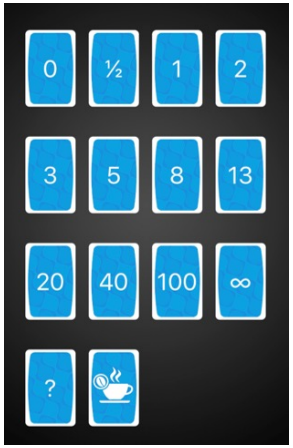
Task Estimation

Estimation

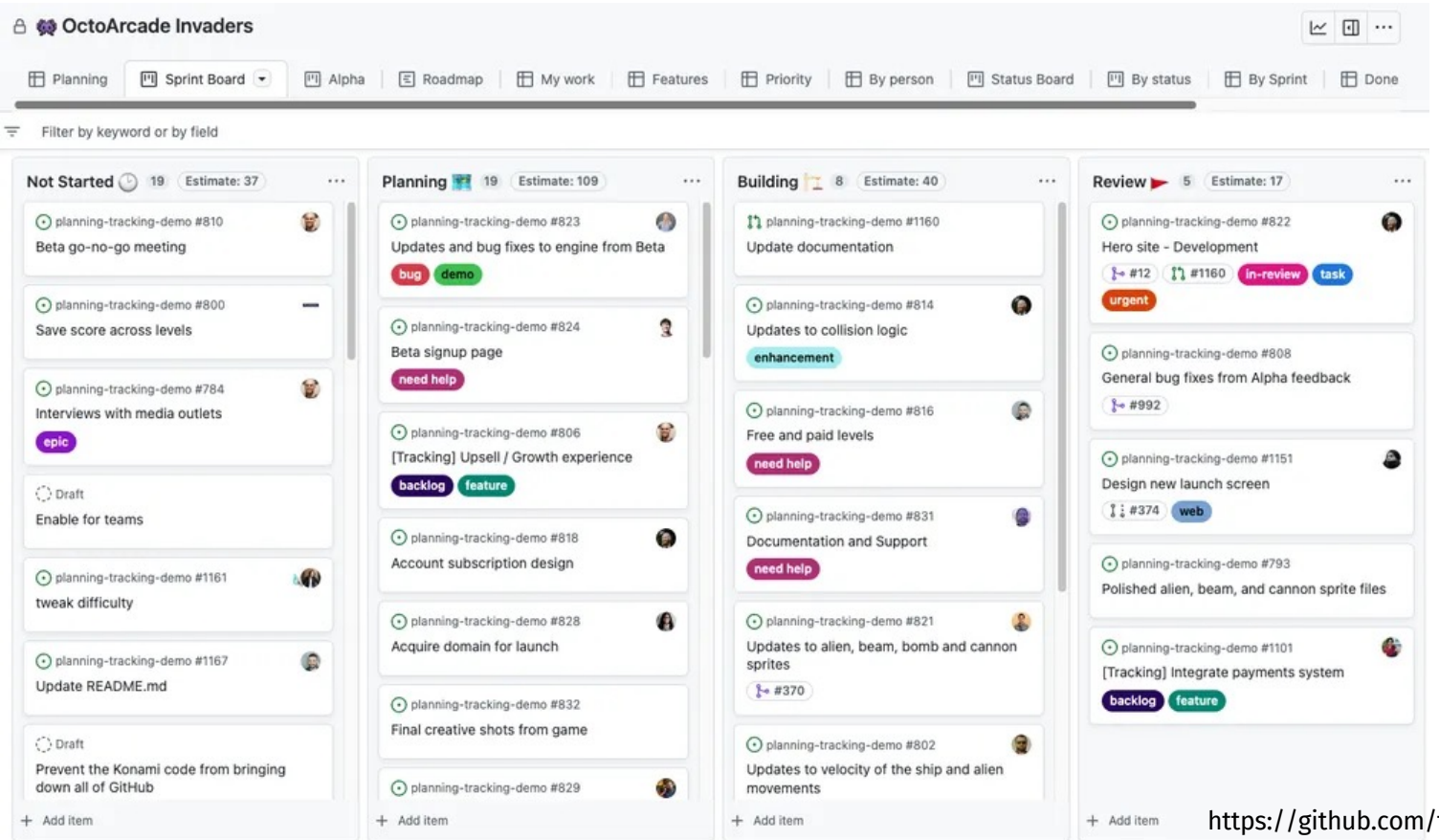
Create a big Lego robot

X Point(s)





Kanban



https://github.com/features/issues

DoR – Definition of Ready (Example)

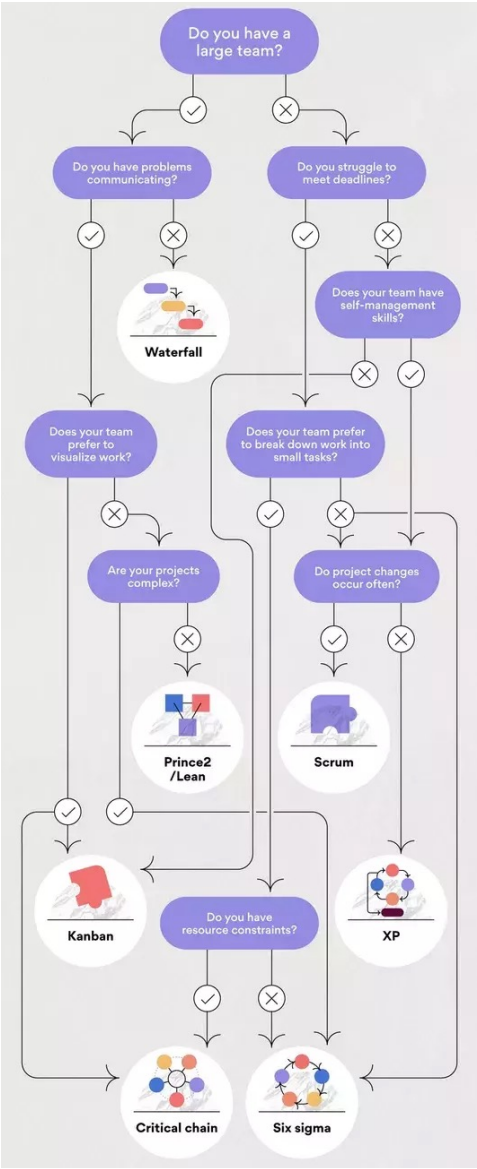
- Story written as a "User Story" (i.e. “As … I want … so that …”)
- Acceptance criteria must exist and be understood by team
- User Story has been estimated by the team (in points)
- Where appropriate: Sketches, diagrams, annexes, reference documents and other resources exist, on a shared access platform
- The User Story is validated by the Product Owner
- The User Story is validated by the Dev Team

DoD – Definition of Done (Example)

- If it is a code development User Story: The code builds and is archived on git
- If it is a code development User Story: The code has been reviewed and merged in a "master branch" on git
- If it is a code development User Story and if applicable: Some automated tests (checking for the User Story's Acceptance Criteria) are running on git
- If it is a code development User Story and if applicable : A documentation is available either as a ReadMe file in the repo or in the general documentation
- If the user story is not code related, All generated document are archived
- All the User Story's Acceptance Criteria are met and verified by the Team (e.g. in the Sprint Review)
- The User Story is accepted by the Product Owner

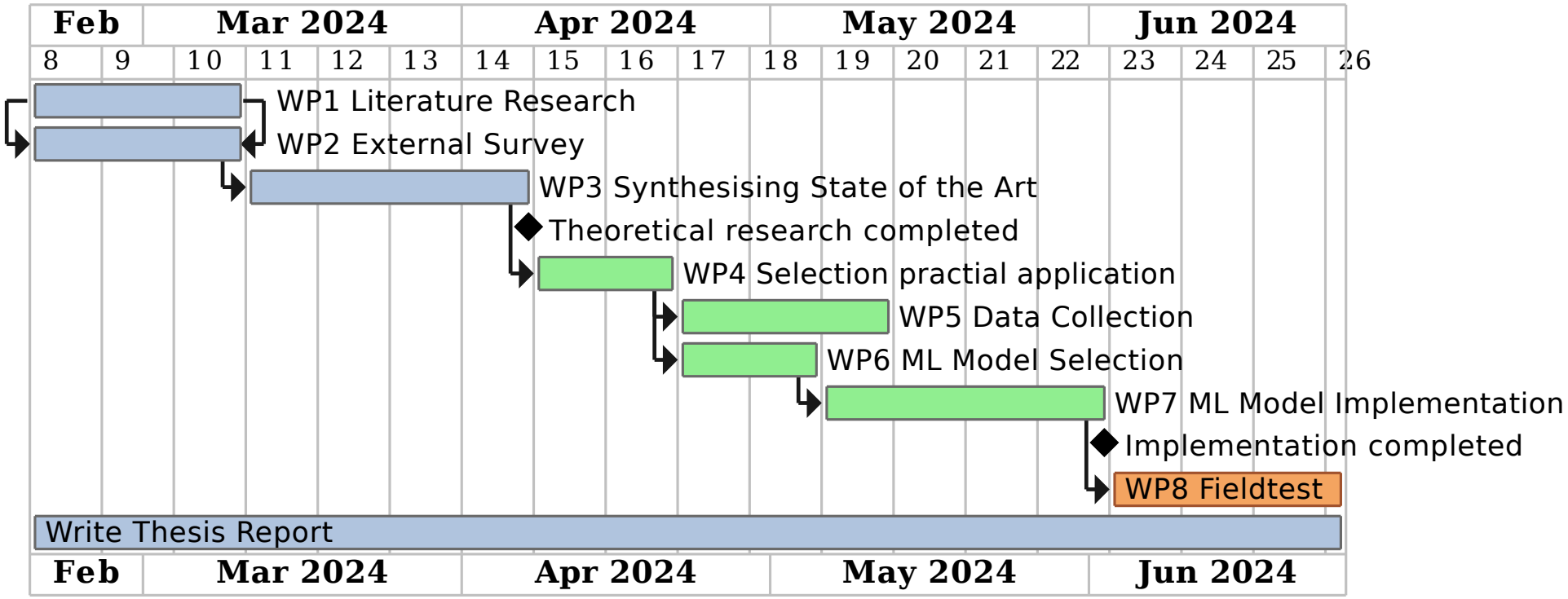
Outro

What is the right method ?

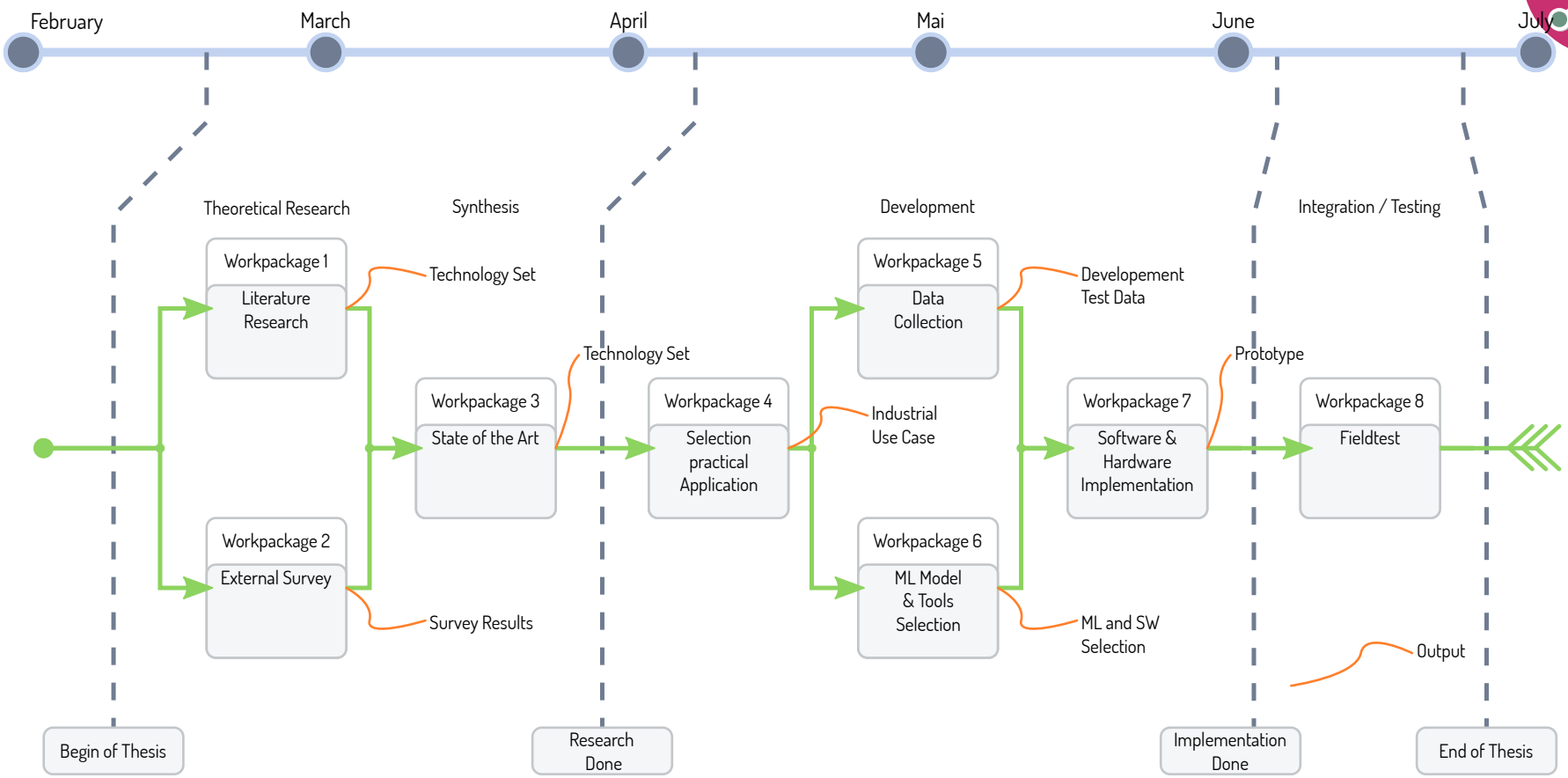


https://asana.com/resources/project-management-methodologies

Gantt-Diagram



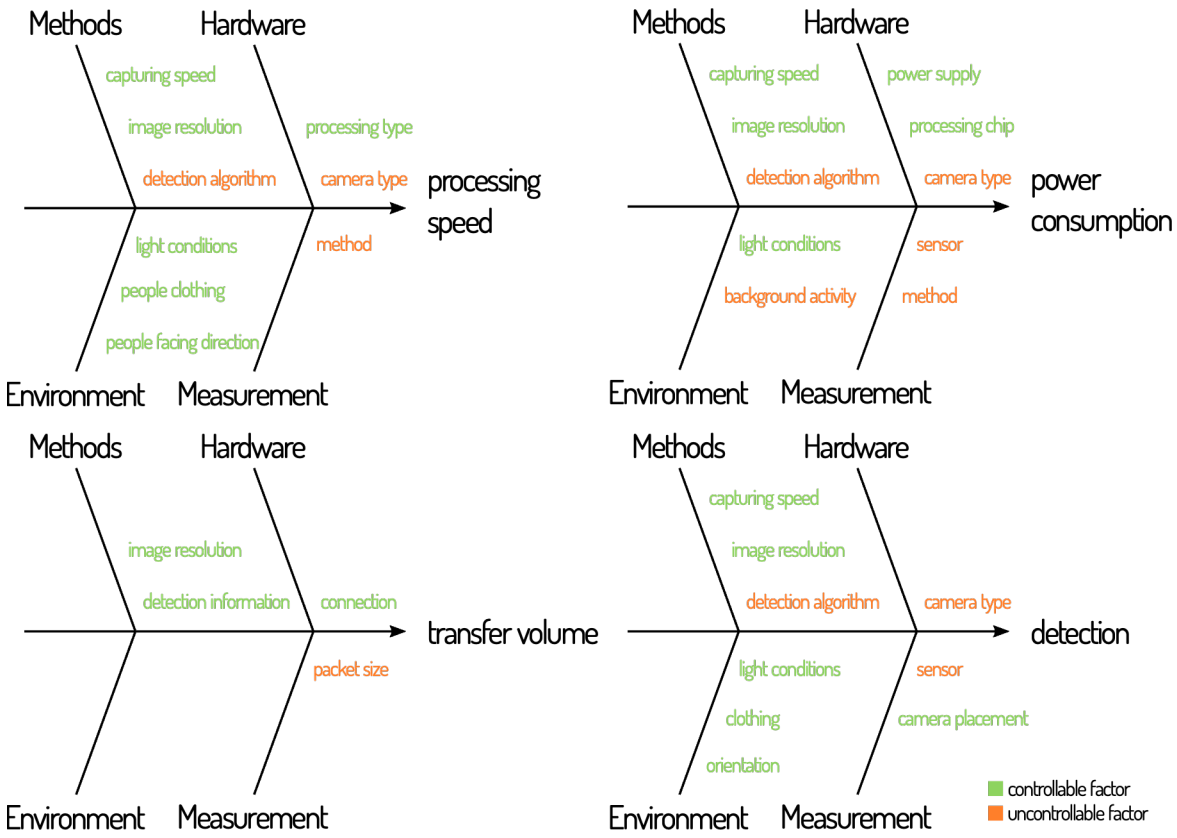
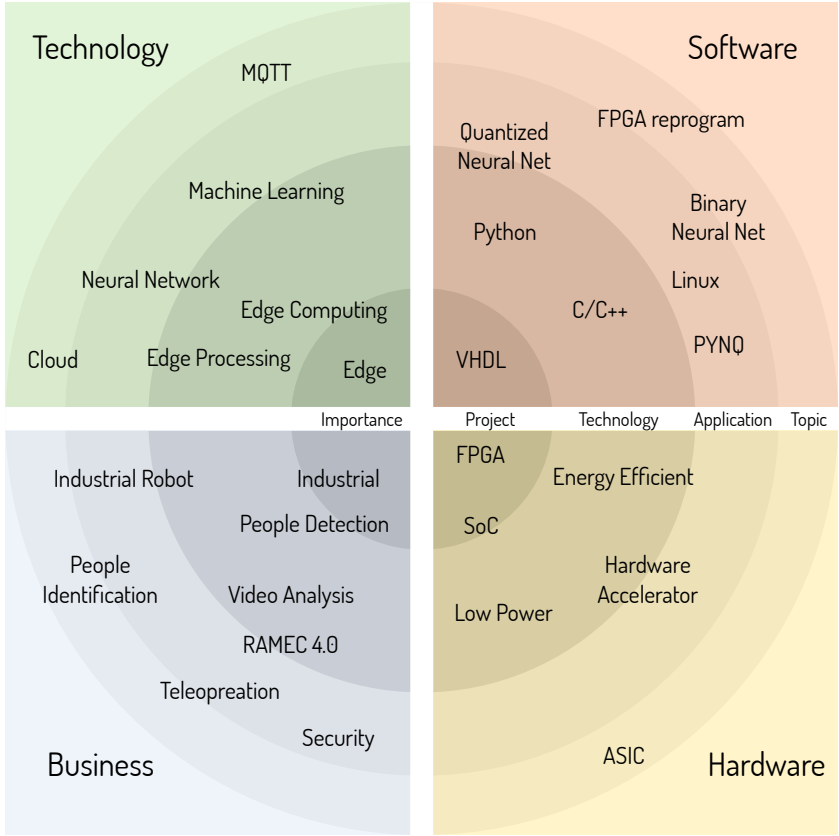
Project Plan



SWOT-Analysis

	Helpful for your objective	Harmful for your objective
Internal within technology	**Strengths** • On-site processing, reducing bandwidth and latency • Longer hardware lifecycle • Realtime capabilities • Adaptive HW for future expansion • Reliable HW for security applications	**Weaknesses** • Many ML nets are implemented and tested on other hardware first • Different programming and HW knowhow required • Custom hardware necessary depending on the application
External outside technology	**Opportunities** • Other types of algorithms can be implemented • Developement of a new ecosystem • Proven technology also used in datacenters, sattelites and medical	**Threats** • CPU and GPU technologies are getting faster and cheaper • User resistance to change to a new technology • Industial sector not used to • Smaller hardware choice available

Cause and Effect Diagram



Keyword Radar